

Vulnerability Assessment & Penetration Testing Report

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6 February 2026

1. Introduction

This report presents the Vulnerability Assessment and Penetration Testing performed on a virtual lab environment using open-source tools such as Kali Linux, Nmap, and Nikto. The objective was to identify security weaknesses in the Metasploitable 2 virtual machine.

2. Objective of Assessment

- To perform vulnerability scanning on a target machine
- To identify misconfigurations and outdated services
- To understand basic VAPT workflow
- To document findings with risk and remediation

3. Tools & Environment

- Kali Linux
- Metasploitable 2
- Nmap
- Nikto
- VirtualBox / VMware

4. Methodology

4.1 Planning

A virtual lab environment was created using Kali Linux and Metasploitable machine.

4.2 Discovery

Nmap was used to discover open ports and running services on the target machine.

4.3 Vulnerability Scanning

Nikto tool was executed to detect web server vulnerabilities.

4.4 Documentation

Findings were recorded with risk levels and remediation.

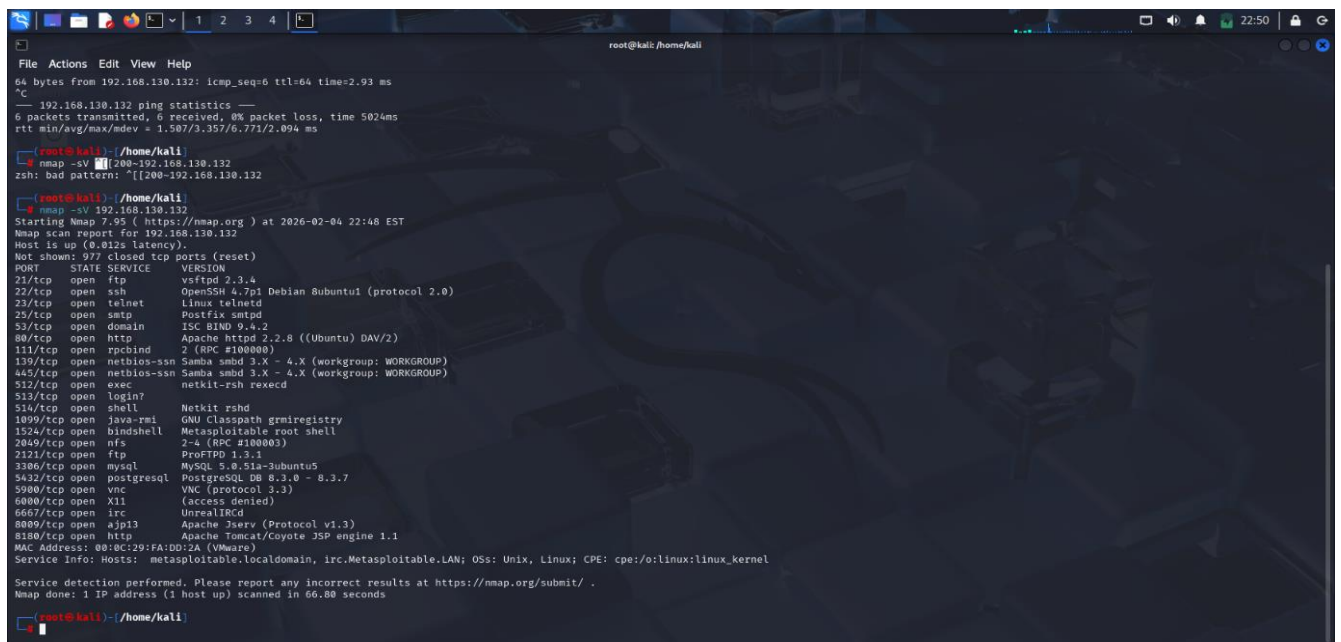
5. Target Details

- Target IP Address: 192.168.130.132
- Environment: Local Virtual Lab

6. Practical Implementation

6.1 Nmap Scan

Nmap was used to identify open ports and services running on the target system.



```
root@kali: /home/kali
64 bytes from 192.168.130.132: icmp_seq=6 ttl=64 time=2.93 ms
^C
--- 192.168.130.132 ping statistics ---
6 packets transmitted, 6 received, 0% packet loss, time 5024ms
rtt min/avg/max/mdev = 1.587/3.357/6.771/2.094 ms
root@kali: /home/kali
nmap -sV 192.168.130.132
zsh: bad pattern: "[[200-192.168.130.132
root@kali: /home/kali
nmap -sV 192.168.130.132
Starting Nmap 7.95 ( https://nmap.org ) at 2026-02-04 22:48 EST
Nmap scan report for 192.168.130.132
Host is up (0.012s latency).
Not shown: 977 closed tcp ports (reset)
PORT      STATE SERVICE      VERSION
21/tcp    open  ftp          vsftpd 2.3.4
22/tcp    open  ssh          OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)
23/tcp    open  telnet       Linux telnetd
25/tcp    open  smtp         Postfix smtpd
52/tcp    open  domain       ISC BIND 9.4.2
80/tcp    open  http         Apache httpd 2.2.8 ((Ubuntu) DAV/2)
111/tcp   open  rpcbind      2 (RPC #100000)
139/tcp   open  netbios-ssn  Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
445/tcp   open  netbios-ssn  Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
512/tcp   open  exec         netkit-rsh rshexec
513/tcp   open  login?
514/tcp   open  shell        Netkit rshd
1099/tcp  open  java-rmi     GNU classpath grmlregistry
1524/tcp  open  bindshell    Metasploitable root shell
2049/tcp  open  nfs          2-4 (RPC #100003)
2121/tcp  open  ftp          ProFTPD 1.3.1
3306/tcp  open  mysql        MySQL 5.0.51a-3ubuntu5
5432/tcp  open  postgresql   PostgreSQL DB 8.3.0 - 8.3.7
5900/tcp  open  vnc          VNC (protocol 3.3)
6000/tcp  open  X11          (access denied)
6667/tcp  open  irc          UnrealIRCd
8009/tcp  open  ajp13        Apache Jserv (Protocol v1.3)
8180/tcp  open  http         Apache Tomcat/Coyote JSP engine 1.1
MAC Address: 08:0C:29:FA:DD:2A (VMware)
Service Info: Hosts: metasploitable.localdomain, irc.Metasploitable.LAN; OS: Unix, Linux; CPE: cpe:/o:linux:linux_kernel
Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 56.88 seconds
root@kali: /home/kali
```

Figure 1: Nmap Scan Output Showing Open Ports and Services

6.2 Nikto Web Vulnerability Scan

Nikto tool was used to scan the web server and detect vulnerabilities such as outdated versions and configuration issues.

```
File Actions Edit View Help
(kali@kali)-[~]
└─$ sudo su
[sudo] password for kali:
(kali@kali) ~ /home/kali
└─$ nikto -h 192.168.130.132
- Nikto v2.5.0

+ Target IP: 192.168.130.132
+ Target Hostname: 192.168.130.132
+ Target Port: 80
+ Start Time: 2026-02-05 03:45:18 (GMT-5)

+ Server: Apache/2.2.8 (Ubuntu) DAV/2
+ /: Retrieved x-powered-by header: PHP/5.2.4-2ubuntu5.10.
+ /: The anti-clickjacking X-Frame-Options header is not present. See: https://developer.mozilla.org/en-US/docs/Web/HTTP/Headers/X-Frame-Options
+ /: The X-Content-Type-Options header is not set. This could allow the user agent to render the content of the site in a different fashion to the MIME type. See: https://www.netsparker.com/web-vulnerability-scanner/vulnerabilities/missing-content-type-header/
+ /index: Uncommon header 'tcn' found, with contents: list.
+ /index: Apache mod_negotiation is enabled with MultiViews, which allows attackers to easily brute force file names. The following alternatives for 'index' were found: index.php. See: http://www.wisec.it/setcou.php?id=46988bdc59d15,https://exchange.xforce.ibmcloud.com/vulnerabilities/8275
+ Apache/2.2.8 appears to be outdated (current is at least Apache/2.4.54). Apache 2.2.34 is the EOL for the 2.x branch.
+ /: Web Server returns a valid response with junk HTTP methods which may cause false positives.
+ /: HTTP TRACE method is active which suggests the host is vulnerable to XST. See: https://owasp.org/www-community/attacks/Cross_Site_Tracing
+ /phpinfo.php: Output from the phpinfo() function was found.
+ /doc/: Directory indexing found.
+ /doc/: The /doc/ directory is browsable. This may be /usr/doc. See: http://cve.mitre.org/cgi-bin/cvename.cgi?name=CVE-1999-0678
+ /?PHPBB85f2A0-3C92-11d3-A3A9-4C7B08C10000: PHP reveals potentially sensitive information via certain HTTP requests that contain specific QUERY strings. See: OSVDB-12184
+ /?PHPPE9568f36-D428-11d2-A769-00AA001ACF42: PHP reveals potentially sensitive information via certain HTTP requests that contain specific QUERY strings. See: OSVDB-12184
+ /?PHPPE9568f34-D428-11d2-A769-00AA001ACF42: PHP reveals potentially sensitive information via certain HTTP requests that contain specific QUERY strings. See: OSVDB-12184
+ /?PHPPE9568f33-D428-11d2-A769-00AA001ACF42: PHP reveals potentially sensitive information via certain HTTP requests that contain specific QUERY strings. See: OSVDB-12184
+ /phpMyAdmin/changelog.php: phpMyAdmin is for managing MySQL databases, and should be protected or limited to authorized hosts.
+ /phpMyAdmin/Changelog: Server may leak inodes via ETags, header found with file /phpMyAdmin/Changelog, inode: 92462, size: 48540, mtime: Tue Dec 9 12:24:00 2008. See: http://cve.mitre.org/cgi-bin/cvename.cgi?name=CVE-2003-1418
+ /phpMyAdmin/Changelog: phpMyAdmin is for managing MySQL databases, and should be protected or limited to authorized hosts.
+ /test/: Directory indexing found.
+ /test/: This might be interesting.
+ /phpinfo.php: PHP is installed, and a test script which runs phpinfo() was found. This gives a lot of system information. See: CWE-552
+ /icons/: Directory indexing found.
+ /icons/README: Apache default file found. See: https://www.vntweb.co.uk/apache-restricting-access-to-iconsreadme/
+ /phpMyAdmin/: phpMyAdmin directory found.
+ /phpMyAdmin/Documentation.html: phpMyAdmin is for managing MySQL databases, and should be protected or limited to authorized hosts.
+ /phpMyAdmin/README: phpMyAdmin is for managing MySQL databases, and should be protected or limited to authorized hosts. See: https://typo3.org/
+ /#wp-config.php#: #wp-config.php# file found. This file contains the credentials.
+ 8910 requests: 0 error(s) and 27 item(s) reported on remote host
+ End Time: 2026-02-05 03:46:04 (GMT-5) (46 seconds)

+ 1 host(s) tested
(kali@kali) ~ /home/kali
└─$
```

Figure 2: Nikto Web Vulnerability Scan Result

6.3 OpenVAS / Greenbone Attempt

OpenVAS vulnerability scanner was installed and configured. However, during execution, multiple technical errors related to SCAP data and feed synchronization were encountered. Despite repeated troubleshooting attempts including reinstallation, feed synchronization, and permission fixes, the dashboard did not function correctly. Therefore, full vulnerability scanning using OpenVAS could not be completed.

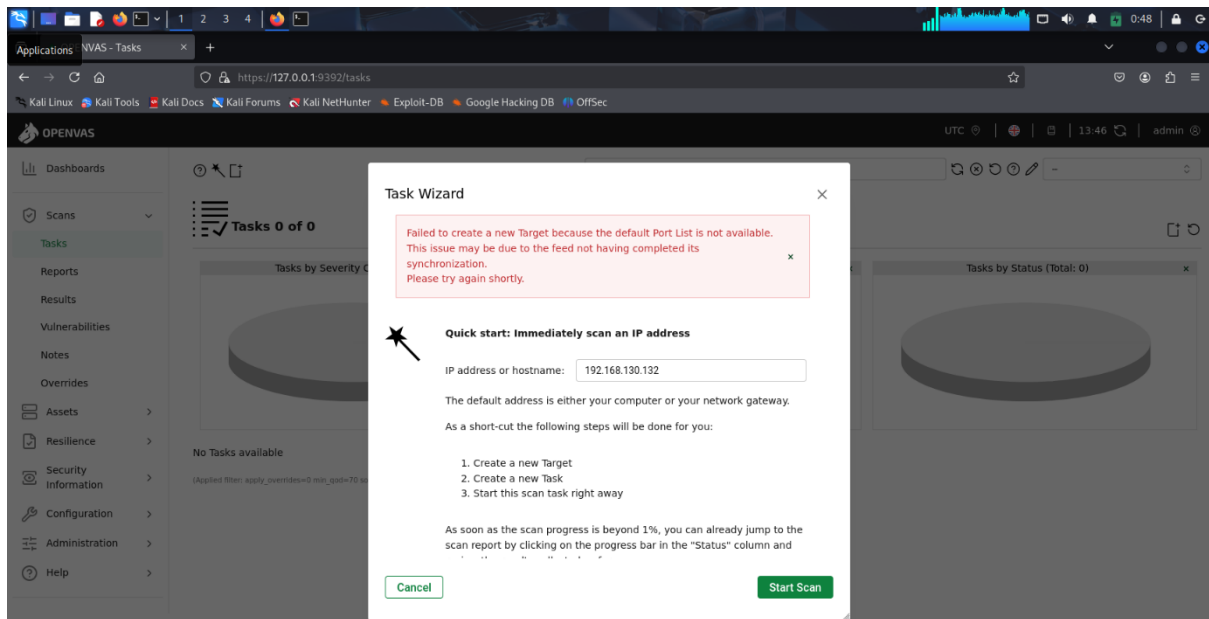


Figure 3: OpenVAS Dashboard Error During Feed Synchronization

7. Vulnerabilities Identified

- Outdated Apache Version
- Directory Indexing Enabled
- phpMyAdmin Exposed
- Missing Security Headers
- phpinfo.php Found

8. Risk Assessment Table

Vulnerability	Risk Level
Outdated Apache Version	High
phpMyAdmin Exposure	High
Directory Indexing Enabled	Medium
Missing Security Headers	Low
phpinfo File Present	Medium

9. Suggested Security Improvements

Based on the vulnerabilities identified during the assessment, the following security improvements are recommended:

- Update Apache to the latest stable version.
- Disable directory listing to prevent unauthorized file access.
- Restrict phpMyAdmin access using authentication or IP controls.
- Add security headers such as X-Frame-Options and X-Content-Type-Options.
- Remove or restrict access to phpinfo.php to avoid information disclosure.

10. Conclusion

The vulnerability assessment successfully identified multiple web server misconfigurations and outdated components using open-source tools. Although OpenVAS scanning faced technical issues, Nmap and Nikto provided sufficient insights into the target system's security posture. The exercise demonstrated practical understanding of basic VAPT workflow and documentation techniques.